

Is the Stability Threshold n_{thresh} Universal?

Scale analysis of Matzke's 6.41-bit threshold

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Abstract

Douglas Matzke's hyperbit framework (IPI 2026) derives a stability threshold

$$n_{\text{thresh}} = \frac{2\pi}{\sqrt{2} \ln 2} = 6.4097 \text{ bits} \quad (.1)$$

and reads it as a universal number: $\text{Cl}(6)$ carries 6.000 bits and thus sits 0.41 bits below the collapse threshold – matter is stable because the smallest fermion algebra just barely fails to collapse. Doc. 326 raised the suspicion that this number hangs on the choice of a particular temperature. This document tests that numerically and settles it.

Main result [B]

With the system-dependent bit energy of FFGFT $E_{\text{bit}}(L) = \hbar c/L$ (Doc. 257) one obtains exactly

$$n_{\text{thresh}}(L) = \frac{L}{\sqrt{2} \ell_{\text{p}}} \quad (.2)$$

– **linear in the scale**, not an invariant number. Matzke's 6.41 corresponds exactly to the scale

$$L_{\text{Matzke}} = \frac{2\pi \ell_P}{\ln 2} = 9.0647 \ell_P. \quad (.3)$$

The "universal" bit value is the FFGFT bit energy, evaluated at about nine Planck lengths. The universality is a concealed choice of scale.

What remains *untouched* is the robust core of Matzke's argument: the collapse condition Compton = Schwarzschild yields $M_{\text{coll}} = m_P/\sqrt{2}$ purely geometrically – without any reference to bits, Landauer, or a temperature. That part stands independently **[B]**.

Verification script: 326_nthresh_skalenabhaengigkeit.py – 12 assertions, all passed.

.1 The question

Doc. 326 works out the central divergence between Matzke's framework and FFGFT: universal vs. system-dependent bit values. The suspicion formulated there was that Matzke's stability threshold "hangs on the postulate $T = T_P$ " and would itself be system-dependent under a system-dependent bit energy – classified as **[S]**.

This question is numerically decidable, and completely so: one recomputes Matzke's chain, extracts the scale-independent parts, replaces the bit value by the FFGFT bit energy, and sees whether a constant remains.

.2 Matzke's chain

The Landauer bridge

The temperature used is not the Planck temperature itself but the **KMS temperature** of the Planck scale:

$$T_{\text{KMS}} = \frac{T_P}{2\pi} = 2.2549 \times 10^{31} \text{ K}. \quad (.4)$$

The 2π comes from the KMS periodicity – the same one that generates the Hawking and Unruh temperatures in Doc. 325. Landauer at this temperature:

$$E_{\text{bit}} = k_B T_{\text{KMS}} \ln 2 \implies m_{\text{bit}} = \frac{m_P \ln 2}{2\pi} = 0.1103 m_P. \quad (.5)$$

Numerically reproduced (script [1], ratio 1.000000000). **[B]**

The collapse condition

Compton wavelength equal to Schwarzschild radius:

$$\frac{\hbar}{Mc} = \frac{2GM}{c^2} \implies M_{\text{coll}} = \sqrt{\frac{\hbar c}{2G}} = \frac{m_{\text{P}}}{\sqrt{2}} = 1.5390 \times 10^{-8} \text{ kg}. \quad (.6)$$

The scale-independent core [B]

$M_{\text{coll}} = m_{\text{P}}/\sqrt{2}$ contains **no bit value**. The collapse condition is purely geometric and identical under any bit interpretation. Numerical check: $n_{\text{thresh}} \cdot m_{\text{bit}} = M_{\text{coll}}$ to nine digits (script [2]).

The threshold

$$n_{\text{thresh}} = \frac{M_{\text{coll}}}{m_{\text{bit}}} = \frac{m_{\text{P}}/\sqrt{2}}{m_{\text{P}} \ln 2 / (2\pi)} = \frac{2\pi}{\sqrt{2} \ln 2} = 6.4097. \quad (.7)$$

The number is thus a *quotient* of a geometric numerator and a posited denominator. Everything about it that is not geometric sits in the bit value.

.3 The same computation with system-dependent bit energy

Instead of a universal bit value, FFGFT posits (Doc. 257, A271–A273):

$$E_{\text{bit}}(L) = \frac{\hbar c}{L} \implies m_{\text{bit}}(L) = \frac{\hbar}{cL}. \quad (.8)$$

Inserted into $n_{\text{thresh}} = M_{\text{coll}}/m_{\text{bit}}$:

$$n_{\text{thresh}}(L) = \frac{m_{\text{P}}}{\sqrt{2}} \cdot \frac{cL}{\hbar} = \frac{L}{\sqrt{2} \ell_{\text{P}}}, \quad (.9)$$

since $\ell_{\text{P}} = \hbar/(cm_{\text{P}})$. Numerically confirmed for $L \in \{1, 5, 10\} \ell_{\text{P}}$ to machine precision (script [3]). **[B]**

What follows

n_{thresh} is a **linear function of the scale**, not a constant. It has no distinguished value as long as no scale is distinguished. The question “how many bits until collapse?” is unanswerable without specifying the scale – it is underdetermined.

Matzke’s implicit scale

Solving $m_{\text{bit}} = \hbar/(cL)$ for L and inserting Matzke’s bit value:

$$L_{\text{Matzke}} = \frac{\hbar}{cm_{\text{bit}}} = \frac{2\pi \ell_{\text{P}}}{\ln 2} = 9.0647 \ell_{\text{P}} = 1.4651 \times 10^{-34} \text{ m}. \quad (.10)$$

And indeed Eq. (.2) returns $n_{\text{thresh}} = 6.4097$ at this scale – exactly Matzke’s value (script [4]). **[B]**

Core finding

Matzke’s universal bit value is the FFGFT bit energy $E_{\text{bit}} = \hbar c/L$, evaluated at $L = 2\pi\ell_P/\ln 2$. Both frameworks compute with the same quantity; the difference is that FFGFT carries the scale along as a free parameter, whereas Matzke fixes one particular scale and calls the result universal.

.4 Numerical range

n_{thresh} at the scales of the corpus (script [5]):

Scale	L [m]	n_{thresh}
$L_0 = \xi\ell_P$ (FFGFT minimal scale, Doc. 180)	2.155×10^{-39}	9.43×10^{-5}
ℓ_P (Planck length)	1.616×10^{-35}	0.7071
$2\pi\ell_P/\ln 2$ (Matzke’s implicit scale)	1.465×10^{-34}	6.4097
1 fm (nuclear physics)	1.000×10^{-15}	4.38×10^{19}
$L_e = \hbar c/E_0$ (Doc. 257)	2.685×10^{-14}	1.17×10^{21}
1 nm (atomic physics)	1.000×10^{-9}	4.38×10^{25}

Range across these scales: a factor of 4.64×10^{29} . A quantity varying over 29 orders of magnitude is not a constant. **[B]**

Sensitivity to the temperature choice

Equivalently, the dependence can be expressed via the posited temperature (script [6]); n_{thresh} scales exactly inversely with it:

Posited temperature	m_{bit}/m_P	n_{thresh}
$0.5 T_{\text{KMS}}$	0.0552	12.82
$0.9 T_{\text{KMS}}$	0.0993	7.12
$1.0 T_{\text{KMS}}$	0.1103	6.41
$1.1 T_{\text{KMS}}$	0.1213	5.83
$2.0 T_{\text{KMS}}$	0.2206	3.20

A deviation of just 10 % in the posited temperature shifts n_{thresh} by 10 % – and thereby moves Cl(6) from 0.41 bits below the threshold to 1.12 bits above it (at $1.1 T_{\text{KMS}}$: $n_{\text{thresh}} = 5.83 < 6.000$). The stability claim flips.

.5 Placement

What falls

The reading of $n_{\text{thresh}} = 6.41$ as a *fundamental number* of the universe. It is the quotient of a geometric quantity and a posited bit value; the postulate is a choice of

scale, and across scales the result varies by 29 orders of magnitude. The statement “Cl(6) lies 0.41 bits below the threshold” holds only at $L = 9.06 \ell_P$ and does not survive beyond it. **[X]**

What stands

Considerably more than the criticism might suggest:

- $M_{\text{coll}} = m_P/\sqrt{2}$ from Compton = Schwarzschild – purely geometric, independent of any bit interpretation **[B]**.
- The Schwarzschild radius without field equations: Matzke’s Novel Result #1 is untouched by this analysis **[B]**.
- Cl(6) as a fermion generation (Furey), the three falling out, the algebraic structure as a whole **[B]**.
- The Hawking mechanism as an algebraic picture – the structural correspondences to FFGFT are exact (Doc. 326).

Consequence for FFGFT

The analysis is not merely a refutation; it supports the FFGFT position of Doc. 257 and A271–A273 from the outside. If a framework that explicitly posits a universal bit value can be reconstructed as the evaluation of the scale-dependent formula $E_{\text{bit}} = \hbar c/L$ at one particular scale, then scale dependence is not the special case but the general form.

This fits with Doc. 325: there the Hawking temperature is derivable without a universal bit value ($k_B T_H = E_{\text{bit}}(4\pi r_s)$ exactly), and the numerical check in Doc. 326 gave $T_H(\text{Matzke})/T_H(\text{KMS}) = 1.000000$. Both findings say the same thing: the universal bit value is a reparametrization, not an independent physical input.

Worth noting is the origin of the 2π in Matzke’s bit value: it comes from the KMS periodicity, i.e. from exactly the relation that generates the Hawking temperature in Doc. 325 *without* a bit value.

.6 Status summary

Statement	Status
$m_{\text{bit}} = m_P \ln 2/(2\pi)$ from Landauer at $T_P/(2\pi)$ reproduced	[B]
$M_{\text{coll}} = m_P/\sqrt{2}$ purely geometric, bit-value free	[B]
$n_{\text{thresh}}(L) = L/(\sqrt{2}\ell_P)$ – linear in the scale	[B]
Matzke’s scale: $L = 2\pi\ell_P/\ln 2 = 9.0647 \ell_P$	[B]
Range across corpus scales: factor 4.6×10^{29}	[B]
$n_{\text{thresh}} \propto 1/T$ – a 10 % temperature error flips the claim	[B]
$n_{\text{thresh}} = 6.41$ as a universal number	[X]
Compton = Schwarzschild, Cl(6) structure, r_s without GR	[B] (untouched)

.7 Proposed register entry R88

R88 – n_{thresh} is scale-dependent; Matzke's scale reconstructed (August 2026): Doc. 329 settles numerically the question booked as **[S]** in Doc. 326. The collapse condition Compton = Schwarzschild yields $M_{\text{coll}} = m_{\text{P}}/\sqrt{2}$ purely geometrically and free of any bit value **[B]**. With the system-dependent bit energy $E_{\text{bit}}(L) = \hbar c/L$ (Doc. 257) one obtains exactly $n_{\text{thresh}}(L) = L/(\sqrt{2}\ell_{\text{P}})$ – linear in the scale **[B]**. Matzke's value 6.4097 corresponds exactly to $L = 2\pi\ell_{\text{P}}/\ln 2 = 9.0647\ell_{\text{P}}$ **[B]**; the "universal" bit value is the FFGFT bit energy at that scale. Across the scales of the corpus n_{thresh} varies by a factor of 4.6×10^{29} ; a 10 % temperature deviation flips the stability claim for Cl(6). Hence $n_{\text{thresh}} = 6.41$ as a universal number is negatively closed **[X]**. Untouched: Compton = Schwarzschild, the Schwarzschild radius without field equations, and the Cl(6) structure **[B]**. Verification script: 326_nthresh_skalenabhaengigkeit.py (12 assertions).

Declared 24 August 2026

Relation to the corpus

Building block	Doc.
Bit energy $E_{\text{bit}} = \hbar c/L$, information unit $\Delta w = 1$	257
Landauer as carrier operation, not abstract information	A271–A273
Elementary cells, bit capacity, partition invariance	302
$\ell_{\text{P}}, m_{\text{P}}, L_0 = \xi\ell_{\text{P}}$	180
KMS relation, $k_{\text{B}}T_{\text{H}} = E_{\text{bit}}(4\pi r_{\text{s}})$	325
Matzke comparison, bit-value divergence	326

References

1. D. Matzke, *Black Holes from Hyperbits*, IPI Annual Conference, August 2026.
2. J. Pascher, Doc. 326 *Black Holes from Hyperbits and from FFGFT*, FFGFT corpus, 2026.
3. J. Pascher, Doc. 325 *The FFGFT Hawking Mechanism*, FFGFT corpus, 2026.
4. J. Pascher, Docs. 257, 302, A271–A273 (bit energy, Landauer), FFGFT corpus.
5. J. Pascher, Doc. 180 *Derivation of L_0* , FFGFT corpus, 2026.
6. R. Landauer, IBM J. Res. Dev. **5**, 183, 1961.

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